SUMMER EXAMINATION 2026

Machine Learning

BSIT - 3rd Semester Time: 90 Minutes Max Marks: 80

Name: ______________ Roll No.: ______________ Date: ______________

1. Machine learning learns patterns from data to make
(A) predictions or decisions (B) only web pages
(C) fixed hand-coded outputs (D) network cables
2. Supervised learning requires
(A) labeled examples (B) only unlabeled data
(C) no target variable (D) random rewards
3. Unsupervised learning searches for
(A) structure without labels (B) known class names
(C) payment status (D) manual rules
4. Reinforcement learning uses actions, states, and
(A) rewards (B) SQL joins
(C) parse trees (D) static labels only
5. The target in a regression problem is usually
(A) continuous (B) a cluster ID only
(C) a token sequence (D) a Boolean rule only
6. A feature is an input variable used by a
(A) model (B) compiler
(C) database index only (D) test runner
7. A training set is used to
(A) fit model parameters (B) choose final policy only
(C) report unbiased test error (D) encrypt records
8. A test set should be held out for
(A) final generalization evaluation (B) feature creation
(C) repeated tuning (D) data collection only
9. Missing numeric values may be imputed with the
(A) median (B) class name only
(C) file path (D) gradient sign
10. An outlier is an observation that is
(A) unusually distant from typical values (B) always incorrect
(C) a missing label (D) a duplicate key only
11. Standardization commonly gives a feature mean 0 and
(A) standard deviation 1 (B) range 1
(C) median 1 (D) variance 0
12. Min-max scaling typically maps values to
(A) a chosen interval such as 0 to 1 (B) integer labels
(C) random signs (D) probabilities always
13. One-hot encoding represents a category with
(A) binary indicator columns (B) a single ordered rank
(C) a text paragraph (D) a distance matrix only
14. Feature selection removes
(A) unhelpful input variables (B) all target labels
(C) test cases (D) model predictions
15. Data leakage occurs when information from the future or target
(A) enters training improperly (B) is normalized
(C) is shuffled (D) is missing
16. Linear regression models a target as a
(A) weighted linear combination (B) tree depth
(C) cluster assignment (D) probability table only
17. Mean squared error averages
(A) squared residuals (B) absolute labels
(C) class probabilities only (D) feature counts
18. Gradient descent moves parameters opposite the
(A) gradient of loss (B) data mean
(C) test set (D) class label
19. The learning rate controls
(A) step size in optimization (B) number of classes
(C) test proportion only (D) feature count
20. Logistic regression commonly outputs
(A) a class probability (B) a cluster centroid
(C) a parse tree (D) a continuous image only
21. The sigmoid function maps real values to
(A) between 0 and 1 (B) between -1 and 1 only
(C) all integers (D) unit vectors only
22. KNN predicts using the labels of nearby
(A) training points (B) decision nodes
(C) principal components only (D) random samples
23. A larger K in KNN usually makes the decision boundary
(A) smoother (B) more jagged
(C) undefined (D) linear always
24. Naive Bayes assumes features are conditionally independent given the
(A) class (B) distance
(C) tree depth (D) reward
25. Bayes theorem relates posterior, likelihood, prior, and
(A) evidence (B) variance only
(C) gradient (D) residual
26. A decision tree splits data using
(A) feature tests (B) random rewards
(C) vector norms only (D) SQL transactions
27. Entropy measures
(A) impurity or uncertainty (B) distance to origin
(C) learning rate (D) number of layers
28. Pruning a tree helps reduce
(A) overfitting (B) missing values
(C) class count (D) feature scaling
29. An SVM seeks a separating hyperplane with a large
(A) margin (B) entropy
(C) learning rate (D) cluster count
30. Support vectors are training points closest to the
(A) decision boundary (B) centroid
(C) root node (D) mean label
31. A kernel can enable SVMs to model
(A) nonlinear boundaries (B) missing values
(C) test leakage (D) class names
32. Bagging trains models on
(A) bootstrap samples (B) one fixed sample only
(C) ordered labels (D) future data
33. Random forests add randomness by selecting subsets of
(A) features at splits (B) test labels
(C) reward values (D) output classes only
34. Boosting builds an ensemble
(A) sequentially focusing on errors (B) in parallel with no interaction
(C) from one tree (D) only from test data
35. Gradient boosting fits new learners to
(A) current residual errors (B) feature names
(C) random labels (D) test predictions only
36. K-means alternates assignment and
(A) centroid update (B) tree pruning
(C) gradient clipping (D) label encoding
37. The K in K-means is the number of
(A) clusters (B) features
(C) iterations only (D) classes in the test set
38. Hierarchical clustering produces a
(A) dendrogram (B) confusion matrix
(C) ROC curve (D) learning curve only
39. DBSCAN identifies dense regions using neighborhood radius and
(A) minimum points (B) tree depth
(C) class weights (D) learning rate
40. PCA finds orthogonal directions ordered by
(A) explained variance (B) class frequency
(C) reward (D) accuracy only
41. The first principal component has the greatest
(A) variance (B) mean label
(C) test error (D) number of samples
42. Accuracy can be misleading when classes are
(A) imbalanced (B) balanced
(C) continuous (D) standardized
43. Precision equals TP divided by
(A) TP plus FP (B) TP plus FN
(C) TN plus FP (D) all negatives
44. Recall equals TP divided by
(A) TP plus FN (B) TP plus FP
(C) TN plus FN (D) all predictions
45. For TP=12 and FN=3, recall is
(A) 0.80 (B) 0.20
(C) 0.75 (D) 0.60
46. Specificity is the true negative rate, TN divided by
(A) TN plus FP (B) TN plus FN
(C) TP plus FP (D) all positives
47. MAE averages the absolute
(A) prediction errors (B) squared features
(C) class labels (D) gradient steps
48. RMSE is the square root of
(A) MSE (B) MAE
(C) R squared (D) variance only
49. R squared measures variance explained relative to a
(A) baseline model (B) confusion matrix
(C) cluster center (D) learning rate
50. Cross-validation repeatedly splits data into training and
(A) validation folds (B) test labels only
(C) features and targets permanently (D) random classes
51. Stratified splitting preserves
(A) class proportions (B) feature means exactly
(C) time order always (D) duplicate rows
52. Overfitting is associated with low training error and high
(A) validation error (B) training accuracy only
(C) bias only (D) regularization strength
53. Underfitting often indicates a model is
(A) too simple (B) too complex
(C) leaking data (D) using too many labels
54. L1 regularization encourages
(A) sparse coefficients (B) large weights
(C) more clusters (D) higher variance
55. L2 regularization penalizes the
(A) sum of squared weights (B) number of classes
(C) absolute residual only (D) test size
56. A hyperparameter is set
(A) before or outside parameter fitting (B) only by backpropagation
(C) by the test label (D) after deployment only
57. Grid search evaluates
(A) specified hyperparameter combinations (B) all possible datasets
(C) only one model (D) random gradients
58. A perceptron is a
(A) single linear threshold unit (B) clustering algorithm
(C) tree ensemble (D) language model
59. An activation function adds
(A) nonlinearity (B) database indexing
(C) data leakage (D) a test split
60. ReLU outputs max of zero and
(A) its input (B) its square
(C) its class (D) its probability
61. Backpropagation computes gradients using the
(A) chain rule (B) Bayes rule only
(C) Markov rule (D) Pythagorean theorem
62. A CNN convolution filter detects
(A) local spatial patterns (B) only class priors
(C) database keys (D) sequence rewards
63. Pooling usually reduces
(A) spatial resolution (B) number of labels
(C) input types (D) training epochs only
64. An RNN processes inputs while maintaining a
(A) hidden state (B) decision stump
(C) centroid (D) confusion table
65. LSTM gates regulate information in the
(A) cell state (B) test set
(C) feature scaler (D) tree root
66. In Q-learning, Q(s,a) estimates expected return for
(A) an action in a state (B) a feature value
(C) a class label (D) a cluster
67. An epsilon-greedy policy explores randomly with probability
(A) epsilon (B) one minus epsilon always
(C) the learning rate only (D) the discount factor only
68. A discount factor near zero emphasizes
(A) immediate rewards (B) distant rewards
(C) feature scaling (D) test accuracy
69. Model drift means the data or relationship changes after
(A) deployment (B) normalization
(C) training split (D) label encoding
70. Monitoring should track performance, latency, and
(A) input or output drift (B) font size
(C) source comments (D) tree names
71. A reproducible pipeline fixes seeds and records
(A) data, code, and configuration (B) only accuracy
(C) only screenshots (D) only labels
72. Fairness evaluation checks whether outcomes differ across
(A) relevant groups (B) random file names
(C) feature columns only (D) epochs only
73. Explainability helps stakeholders understand
(A) model predictions (B) network cables
(C) database backups (D) file compression
74. A 2 by 2 confusion matrix contains
(A) four outcome cells (B) two models
(C) two features (D) four classes necessarily
75. If a classifier gets 90 correct out of 100, accuracy is
(A) 0.90 (B) 0.10
(C) 0.50 (D) 90.0
76. A lower validation loss generally indicates
(A) better validation fit (B) more leakage necessarily
(C) higher bias always (D) fewer samples only
77. The residual in regression is actual value minus
(A) predicted value (B) feature mean
(C) class prior (D) tree depth
78. The ROC AUC summarizes ranking quality across
(A) classification thresholds (B) learning rates only
(C) cluster counts (D) feature encodings
79. Early stopping can reduce overfitting by halting when validation performance
(A) stops improving (B) reaches zero training loss
(C) has more features (D) uses no labels
80. An epoch is one pass through the
(A) training data (B) test labels only
(C) hyperparameter grid (D) confusion matrix